Chapter 1
Developing a Detection Algorithm

1.1 Detection Algorithm Metrics

A successful detection algorithm requires maximizing the ability to successfully detect the presence of an insect in the store product material sample while minimizing the possibility of falsly identifying a pest in a clean sample and failing to detect an insect in an infested sample. The binary form of the detection algorithm is defined as a decision, D, where $D = 1$ indicates the presence of an insect and $D = 0$ indicates the absence of an insect. Equation 1.1 shows the binary detection algorithm:

$$D = \begin{cases} 1 & \text{Insect Detected} \\ 0 & \text{No Insect Detected} \end{cases} \tag{1.1}$$

The probability of detection for an algorithm is measured by the number of times the algorithm correctly identified the presense of an insect in the stored product material sample divided by the total number of opportunities where an insect was present in the tested material. Equation 1.2 shows the calculation of the probability of detection:

$$\text{PD} = \frac{N(D = 1, I = 1)}{N(I = 1)} \tag{1.2}$$

where $N(D = 1, I = 1)$ is the number of records where the algorithm correctly identified the presence of an insect and $N(I = 1)$ is the total number of records available with insects present in the tested material.

1.2 Normalized Signal Pulse Amplitude as a Criteria for Insect Detection

The NSPA metric demonstrated a greater range of values across the different insect-material pairings compared to the NSEL metric. This suggests that the NSPA may be more sensitive to variations in insect size and activity level, making it a potentially more effective parameter for insect detection applications. A statistical variable, V, can be used as a detection criteria when it exceeds a definite threshold, T. If $V > T$, then an insect is detected, otherwise no insect is detected.

The NSPA measurements, V, from the different insect-material pairings ranged from 26.5 dB to 74.8 dB when using the frequency range 500 Hz to 6000 Hz for filtering and normalization. This means that all insects are detected when T is set to the standard 6 dB popular in other SNR based detection methods, such as the SONAR equation [**rudmoseCaseMissing61982**, **walreeUUVCOVERTACOUSTIC**, **peyvandiSONARSystemsUnderwater2011**].

1.3 Generation of Insect-like Signals by External Noise

The developed acoustic insect detection system provided sufficient physical soundproofing that enabled measurement of insect signatures in laboratory conditions with high SNR by the integrated piezoelectric sensors without interference from external noise.

Strong external noise with impulse characteristics, however, could still penetrate inside the enclosure and generate signals similar to those produced by insects and mask their presence. This is highlighted in Figure 1.1 which shows the spectrograms of the recordings from the piezoelectric sensors, internal microphones, and external microphone during a recording of amplified speech at 90 dBA. The amplitude color scales of the piezoelectric and microphone spectrograms are adjusted and differentiated to highlight the impulses generated by the external sound.

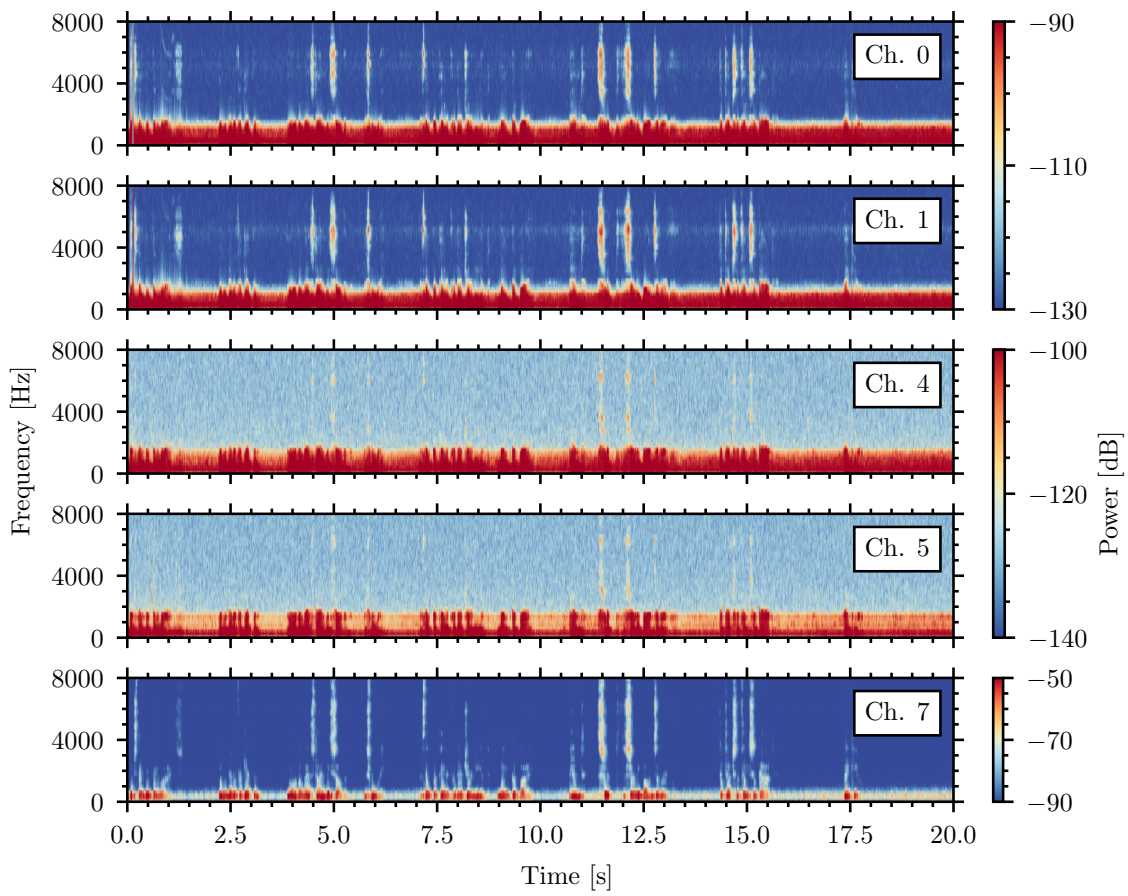

Figure 1.1: Spectrogram from piezoelectric and acoustic microphones during external noise, amplified speech at 90 dBA. Insect like pulses are visible in all channels

Various kinds of noise can produce different peak levels for the same averaged sound level. The insect-like signals generated by the noise peaks, as seen in Figure 1.1, would result in more false alarms than monotonic noise. The crest factor parameter characterizes the impulsive nature of the signal through its peak-to-RMS ratio [**kowalska-koczwaraInfluenceCrestFactor2018**] Equation 1.3 shows the calculation of the crest factor, CR, in decibels for a signal V:

$$\text{Crest Factor} = 20\log_{10}(\frac{V_{\text{peak}}}{V_{\text{RMS}}}) \tag{1.3}$$

The expected noises at U.S. Ports of Entry, such as machinery and vehicles, generate continuous signals with low crest factors ranging from 1 dB to 3 dB [**starckIndustrialImpulseNoise1987**, **bergerNoiseManual2003**], and impulsive noises such as hammering and pneumatic tools with higher crest factors ranging from 5 dB to 10 dB [**starckIndustrialImpulseNoise1987**]. The majority of the industrial noise also has lower frequency content [**salomonsPracticalRangesLoudnes** and the significant part of its energy would be suppressed through filtering in the developed algorithm. Analysis of the recorded signals, after normalized filtered envelope processing, of factory noise [**viertelnachvierFactory22014**, **inspectorjAmbienceMachineFactory2017**] and male speech showed a crest factor of 8 dB and 13 dB, respectively. This demonstrated that the high crest factor speech recordings would be more disruptive to the insect detection algorithm than the industrial noise at the same sound level. If the developed algorithm allows insect detection at noise levels of 100 dBA for the speech noise recording, it would be expected to perform well in the presence of industrial noise even at higher sound levels.

The spectrograms show the variation of the external signal as it penetrates to the piezoceramic sensors and internal microphones. The external microphone, channel seven, shows the signal in air and highly exceeds the magnitude levels compared to the internal sensors. The impulses are similar to the insect signals shown in Figure **??** and **??**. The NSPA values calculated from the noise recordings range from 39.3 dB to 69.4 dB at noise levels from 60 dBA to 100 dBA, respectively, which are comparable or exceed those of the insect signals. This demonstrates that the detection algorithm based on NSPA will report a false positive, an insect detection when no insect is present. Figure 1.2 visualizes the NSPA values calculated from the internal piezoelectric sensors, during the external speech noise at different sound levels against the NSPA values from the different insect-material pairings previously shown in Figure **??**.

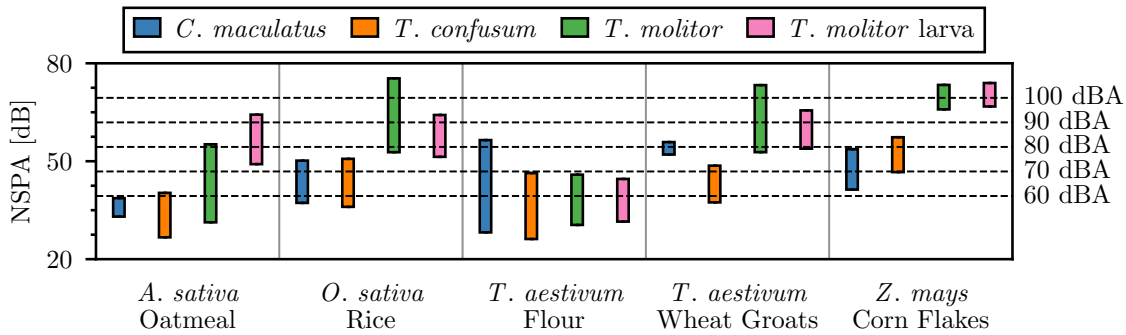

Figure 1.2: NSPA values from internal piezoelectric sensors during external speech noise at different sound levels compared to NSPA values from different insect-material pairings

In Figure 1.1, pulses observed by the internal microphones, channels four and five, are approximately 50 dB lower than those recorded by external microphone seven. This large drop in the acoustic signal magnitude within the soundproofed chamber demonstrates the effectiveness of the sound reduction of the system. Figure 1.3 presents the spectra of an impulse with the maximal amplitude calculated in the short time window of 100 ms for acoustic microphones four, five, and seven in the presence of the external speech noise with a level of 90 dBA. The reference signal of the averaged self-noise from the internal microphones, channels four and five, is also shown.

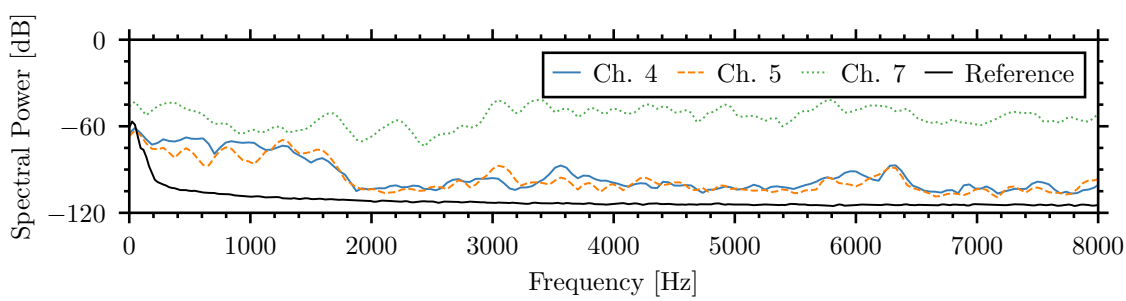

Figure 1.3: Spectra of strong pulses recorded by internal microphone, channel four and five, and external microphone, channel seven,during 90 dBA external noise

The difference in the spectral power between the microphones demonstrates the effectiveness of the insulation in muffling external noise especially at frequencies above 2000 Hz where it was able to provide an average of 45.4 dB and a maximum of 58.9 dB noise reduction. This level of attenuation is comparable to the results achieved with the 620 kg chamber used for the InsectSound100 dataset but in a more compact and portable form factor [**brandingNoiseRobustAcoustic2023**, **brandingInsectSound1000InsectSound2024**].

The high sensitivity of the piezoelectric discs makes them vulnerable to external noise even through the insulation. The filtered envelope of the normalized signal can visualize the waveforms of the acoustic recordings from the system. Figure 1.4 shows the waveforms, filtered, enveloped, and normalized in the 500 Hz to 6000 Hz frequency range, for two examples of weak and medium insect signals, the *T. confusum* in flour and the *C. maculatus* in wheat groats, respectively, as well as the speech noise at 60 dBA and 80 dBA. The graphics are accompanied by the NSPA values calculated over the same frequency range.

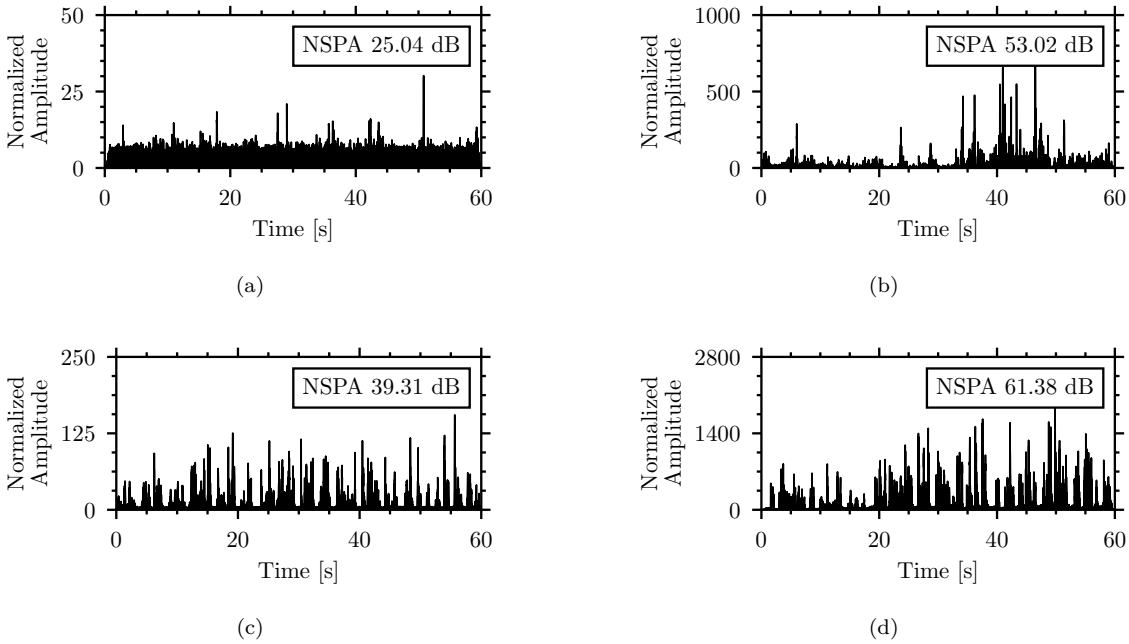

Figure 1.4: Filtered, enveloped, and normalized waveforms (500 Hz to 6000 Hz) of (a) weak strength insect signal of *T. confusum* in flour, (b) medium insect signal of *C. maculatus* in wheat groats, (c) signal of 60 dBA external speech noise, (d) signal of 90 dBA external speech noise. The graphics are accompanied by their NSPA value calculated in the same frequency range.

The waveforms show that the generated pulses from the external noise reach similar amplitudes and duration to those from insect activity. Even at 60 dBA, in Figure 1.4c, the impulse amplitude and NSPA highly exceed the those in Figure 1.4a of the *T. confusum* in flour. The medium strength insect signal, in Figure 1.4b, has a lower amplitude and NSPA compared to the noise at 90 dBA in Figure 1.4d. This further demonstrates that the NSPA based detection algorithm in the considered form is not applicable here since the external noise can generate insect-like signals that exceed those from actual insect activity.

Figure 1.5 compares the spectra, averaged over 1 min, recorded by the internal piezoelectric sensor of the weak and medium signals with the speech noise at 90 dBA. While external noise exhibits dominant spectral power below 1600 Hz, insect signals demonstrate higher energy content above this frequency.

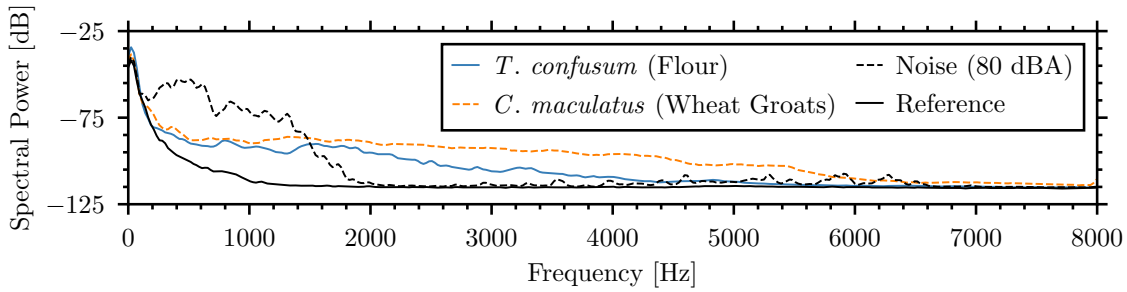

Figure 1.5: Spectral power of *T. confusum* in flour, *C. maculatus* in wheat groats, and external speech noise at 90 dB recorded by channel zero

These spectra show that noise highly exceeds insect signal in the frequencies below 1500 Hz but insect signals have higher energy content at higher frequencies. Adjusting the frequency band of the system could decrease the influence of external noise on the piezoelectric sensors. This adjustment was made based on the frequency dependence of the NSPA algorithm.

The lower and upper frequency limits of the NSPA algorithm were parametrized in order to optimize the differentiation between insect and noise signals. Twenty samples of the insect signals representative of the median strength for each respective insect-material pairing were selected for analysis. The frequency bands were systematically varied and the respective NSPA values were calculated and compared to the NSPA values of the external noise at 90 dBA to maximize the difference.

The NSPA difference significantly increased when the lower frequency of the band-pass filter was above 1500 Hz. The upper frequency limit plays a less critical role. Focusing on the lower frequency range, Figure 1.6 shows the NSPA difference between the insect and noise signals as the lower frequency bound is varied from 1000 Hz to 5000 Hz while the upper frequency bound is held constant at 6000 Hz.

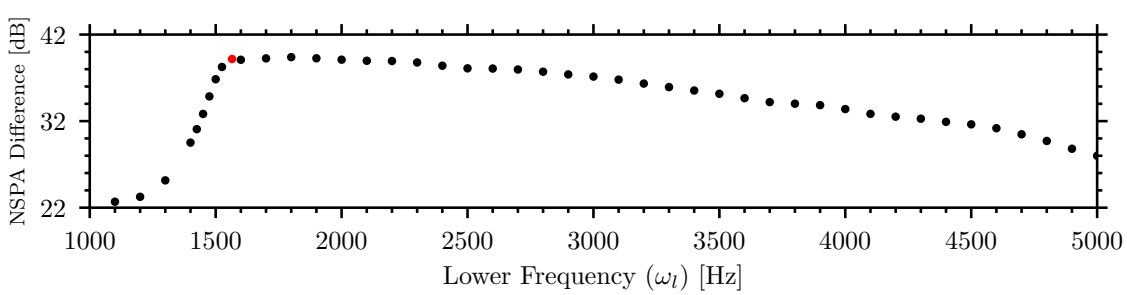

Figure 1.6: Dependence of NSPA difference between insect and external noise signals to the lower frequency of the band-pass filter

The optimal low frequency, when the difference between the insect and noise NSPA is maximized, converged to 1565 Hz before it started to decrease as a result of losing insect signal energy at higher frequencies. The frequency range for filtering and normalization was adjusted to 1565 Hz to 6000 Hz.

Figure 1.7 shows the filtered and normalized waveforms for the same recordings as Figure 1.4 but using the updated frequency range:

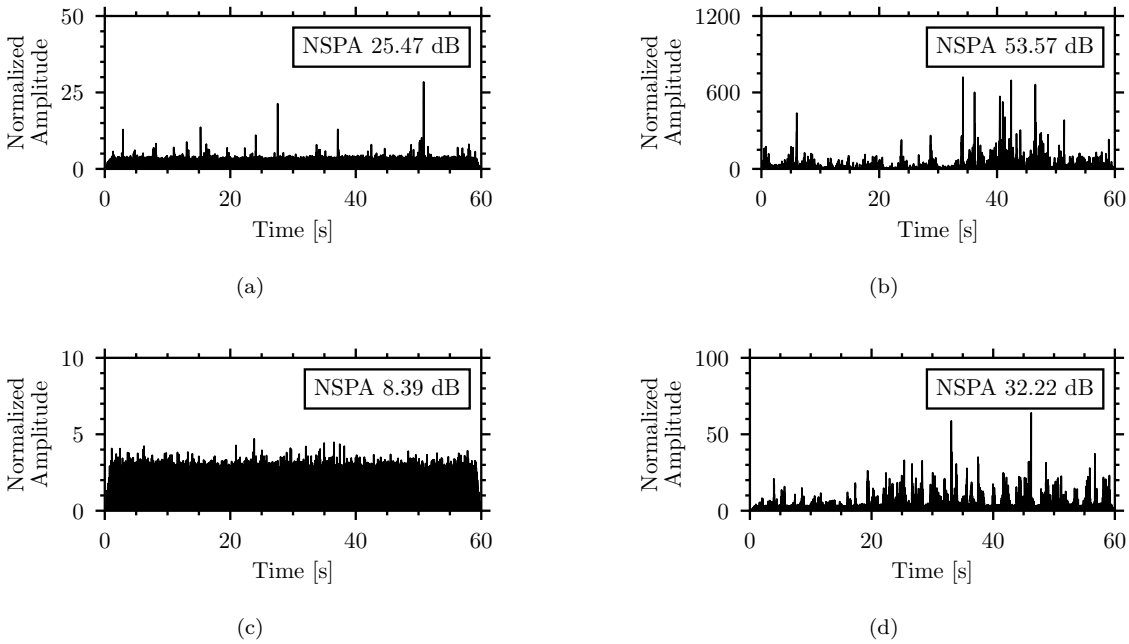

Figure 1.7: Normalized envelope of signal filtered in band 1565 Hz to 6000 Hz recorded by the internal piezoelectric of (a) weak strength *T. confusum* in flour, (b) medium strength *C. maculatus* in wheat groats, (c) 60 dBA external speech noise, and (d) 90 dBA external speech noise

Comparison of Figures 1.4 and 1.7 shows that adjustment of the low cutoff frequency significantly decreases the NSPA produced by external noise.

The NSPA values of the different insect-material pairings and external noise recordings were calculated using the frequency range of 1565 Hz to 6000 Hz. Figure 1.8 shows the distribution of the NSPA values using the adjusted frequency range. Accompanying the box plot are horizontal lines representing the NSPA values measured by the piezoelectric sensors at different external noise levels from 60 dBA to 100 dBA.

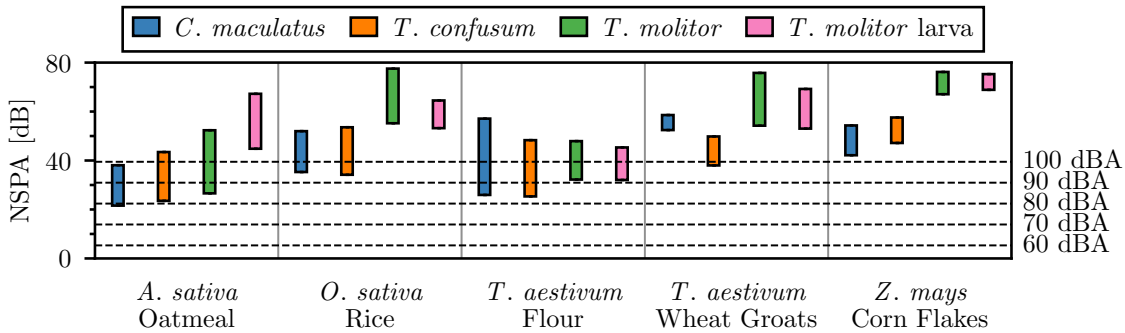

Figure 1.8: Distribution of NSPA values using 1565 Hz to 6000 Hz for different insect-material pairings and external noise

The updated frequency range significantly improved the NSPA difference between the insect signals and the external noise. At a noise level of 80 dBA, nearly all measured NSPA values exceed those generated by external noise. The resulting NSPA range of the insect-material pairings now extends from 21.6 dB to 77.5 dB. If the threshold, T, is set to the maximum NSPA value of the external noise at 80 dBA, 22.4 dB, 99.4 % of the insect-material pairings would be correctly identified as containing an insect. At this threshold, any external noise above 80 dBA could generate false positive insect detections. Increasing the threshold to 30.9 dB, the maximum NSPA value of the external noise at 90 dBA, would result in 90.2 % correct insect detections.

1.4 Leveraging External Acoustic Microphone for Noise Control

The previous sections demonstrated that even single small insects can be detected by the developed system using the NSPA threshold algorithm under speech noise sound levels up to 80 dBA. Additional level of noise protection can be achieved by using noise signal recording by external microphone. The suggested external microphone noise control can allow insect detection in noisy condition and can be used for insect detection in large volumes of products where the insect signals could be small due to longer propagation paths.

A coefficient, K, can be defined to describe the penetration of the external sound to the piezoelectric sensor, representing the ratio between the insect-like signals recorded by the internal piezoelectric sensors and the signal from the external microphone. The average value of the penetration coefficient, K, was found to be approximately 0.03, or -31 dB, across the four channels and five external noise levels from 60 dBA to 90 dBA. The microphone signal can be scaled by this coefficient to estimate the signal on the internal piezoelectric sensors generated by the external noise, $V_{\text{noise}} = KV_{\text{mic}}$. When the measured NSPA(V) is greater than both NSPA(KV_{mic}) and the threshold, T, external noise does not influence insect detection. But if the estimated noise signal exceeds the threshold, NSPA(KV_{mic}) $> T$, the ambient noise can generate a false positive detection even when no insect is present and a noise suppression algorithm can be applied.

The noise suppression algorithm identifies short time windows where external noise generates signals with amplitudes that are close to or exceed the threshold amplitude and can affect the NSPA measurement. Equation 1.4 describes the amplitude threshold limit:

$$A_T = 10^{\frac{T}{20}} \tag{1.4}$$

Peaks in the microphone signal with an amplitude, V, that exceed $A_T/2K$ are identified. A time window of 10 ms centered around each peak is used to search for corresponding peaks in the piezoelectric sensor waveforms. If a peak is found within this time window, it is zeroed out. The NSPA calculation is then applied after the noise removal process.

Figure 1.9 demonstrates the application of the algorithm with the 90 dBA external speech noise shown in the previous figures combined with a recording of a relatively weaker insect signal of *T. confusum* in rice. Figure 1.9a shows the insect waveform, before adding noise, with an NSPA of 30.19 dB and six identified impulse peaks. If an NSPA threshold, T, higher than this value would have been chosen to accommodate for the external noise, the insect signal would not have been detected, resulting in a false negative. Figure 1.9b shows the combined waveform of the external noise and insect signal, with a higher NSPA and 25 impulse peaks. Figure 1.9c shows the cleaned waveform after applying the noise removal algorithm. The NSPA of the cleaned waveform is 30.26 dB with the six original insect peaks preserved while the 19 additional noise generated peaks have been removed.

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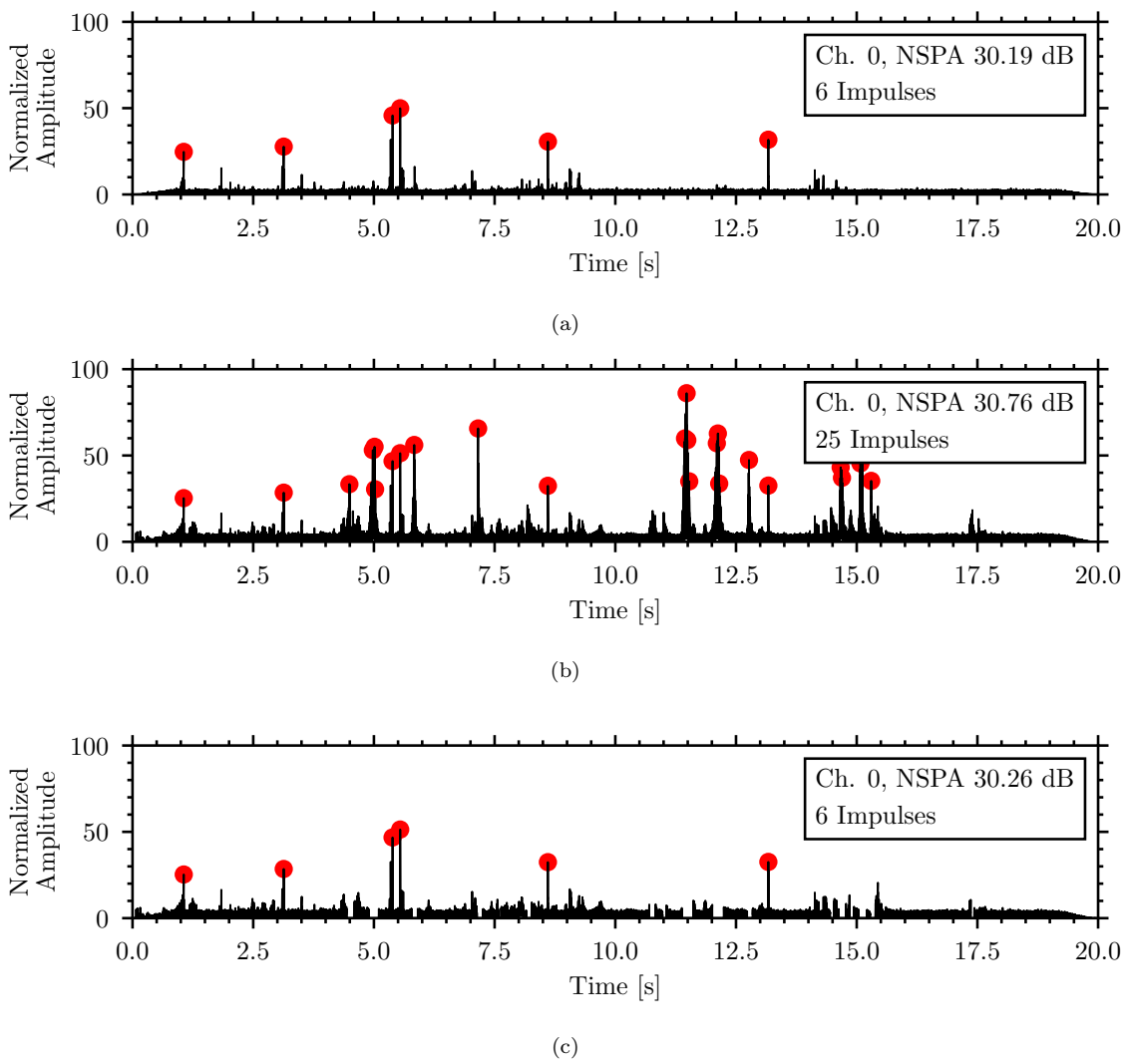

Figure 1.9: Simulation of external noise algorithm with 90 dBA speech noise and *T. confusum* in flour (a) insect signal, (b) combined insect signal and external noise, (c) combined signal after algorithm.

The second example illustrates the algorithm with a 5 s recording of a chainsaw, representing a high sound level scenario that would be encountered as machinery noise at inspection facilities, retrieved from the EC-50 Environmental Sound Classification Dataset [**piczakESCDatasetEnvironmental2015**]. The recording was adjusted by a ratio of its RMS compared to the RMS of the 90 dBA speech noise recording to represent acquisition by the external microphone, channel seven. A copy of the recording was suppressed by the penetration coefficient, K, to simulate the piezoelectric sensor, channel zero.

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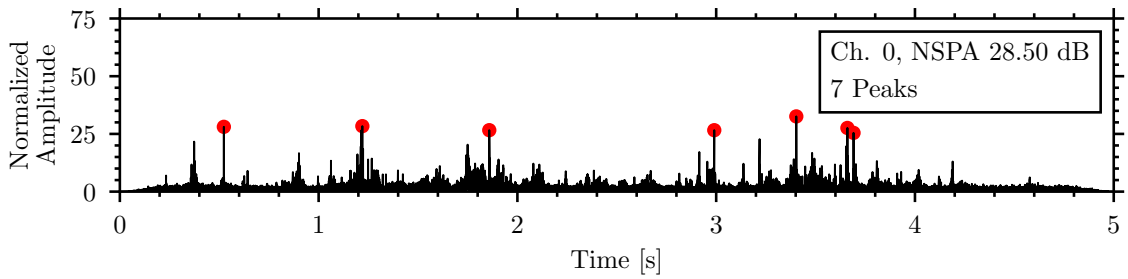
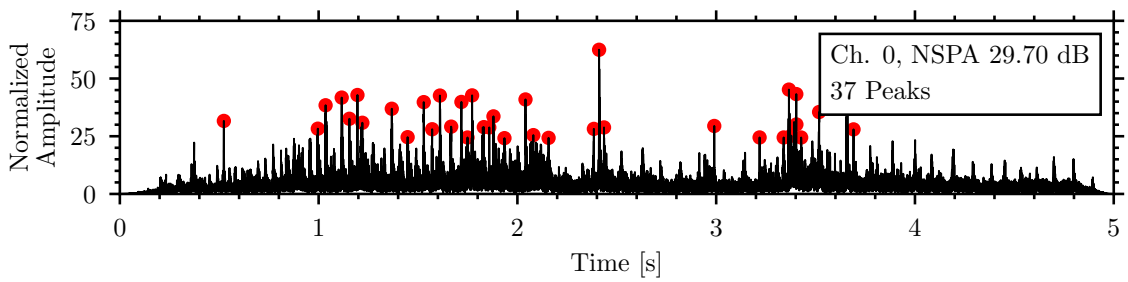
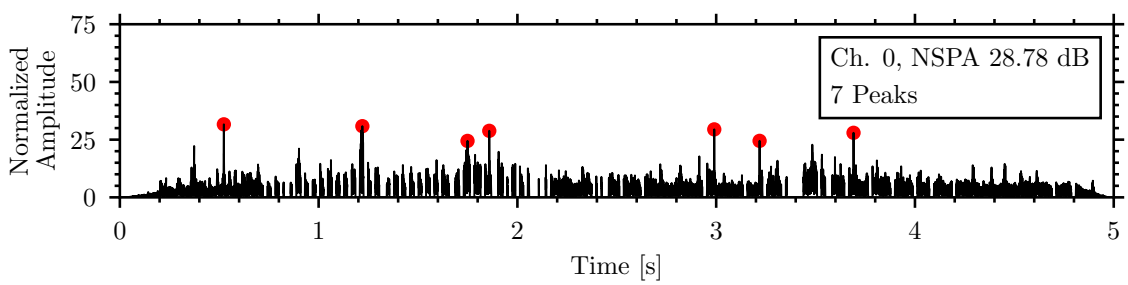


(a) Insect signal before adding noise



(b) Combined waveform of external noise and insect signal



(c) Cleaned combined waveform after applying the noise removal algorithm

Figure 1.10: Simulation of external noise algorithm with chainsaw noise from EC-50 Dataset and *T. molitor* in oatmeal

1.5 Proposed Inspection Procedure and Finalized Detection Algorithm

Deployment of the A-SPIDS platform can follow the proposed subsequent procedure to ensure reliable insect detection even in high external noise environments.

It is expected that the frequency band for filtering and normalization, f_1 and f_2, and the insect detection thresholds, T, are established based on preliminary tests with insects expected to be found in the inspected materials. The data presented in this thesis can be used as a reference for setting these parameters.

At the start of inspection operations, the system is calibrated by recording several minutes of ambient noise without insects present to establish the normalization baseline and the penetration coefficient, K, specific to the environment.

The operator places the sampled material into the container and after securing the lid, initiates the recording process. After a minute, during which the material settles, the recording session begins and continues for a predetermined duration, typically between one to five minutes, depending on the inspection requirements. The signals from the internal piezoelectric sensors, channels zero to three, and the external microphone, channel seven, are processed in real-time during which they are filtered, enveloped, and normalized from f_1 to f_2. If impulses in the external microphone exceed the threshold, $\frac{A_T}{2K}$, the noise removal algorithm is applied to the internal piezoelectric sensor recordings. At one minute segments, the NSPA is calculated. Figure 1.11 diagrams the signal processing up to the insect detection decision:

Figure 1.11: Flowchart of the signal processing including the external noise removal algorithm and insect detection procedure. The microphone signal procesing is marked in blue.

The detection decision for each piezoelectric channel is made based on the NSPA value and the impulse count, N, after noise removal. If both criteria, NSPA $\geq T$ and $N \geq N_{\text{min}}$, are satisfied, an insect is detected on that channel. If either criterion is not met, no insect is detected on that channel. The overall detection decision is made if any of the piezoelectric channels indicate the presence of an insect.